WHICH FEATURE NETWORK MAKES A GOOD PERCEPTUAL LOSS FOR LOW-DOSE CT DENOISING?

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ABSTRACT

Perceptual losses, usually built on ImageNet VGG features, reduce the over-smoothing of low-dose CT (LDCT) denoisers trained with pixel losses. Whether larger, self-supervised or medical vision-language networks make better perceptual losses for CT has not been tested under one protocol. We compare ten perceptual-loss configurations, seven pretrained feature networks from five families and a small CT dose network, with pointwise or local statistics matching, as losses for one U-Net with matched loss influence and three seeds, and test a training-free dose and blur screen. Every perceptual loss lowers LPIPS but costs PSNR against L1; the best trade-offs nearly halve LPIPS (0.104 to about 0.058) at 0.5 to 0.7 dB, with no clear winner. Statistics matching lowers the DINOv2-S texture error from 0.333 to 0.195 ($p = 4.9 \times 10^{-4}$, $r = -0.77$) at no detectable PSNR cost. The screen does not predict the trained outcome, and no larger, self-supervised or medical network clearly beats the default VGG.

***Index Terms*—** Low-dose CT, denoising, perceptual loss, foundation models, noise texture

1. INTRODUCTION

In low-dose CT (LDCT) denoising, pixel losses such as L1 over-smooth the output and erase the noise texture that radiologists read. Perceptual losses are the common fix: the denoiser learns to match the features of the normal-dose image in a pretrained network, which keeps the texture [1].

VGG features are the default perceptual loss [1, 2]. Features of CT-specific networks [3], language-aligned models [4] and foundation models [5, 6] have since appeared, each inside a single method and never compared under one protocol.

A larger or more specialised network is not obviously better. Better ImageNet classifiers are not necessarily better perceptual similarity metrics [7], and features from a domain-specific foundation model did not improve retinal image synthesis [8]. A large medical vision-language model already scores LDCT quality in close agreement with radiologists [9], but a feature network may also reward blur, so our screen tests for it directly.

The closest study [10] varies only the VGG-19 layer and its pretraining set, with one seed and pixel metrics (PSNR, SSIM, NRMSE). We keep its loss-weight calibration and compare networks of different families and sizes, with pointwise or local statistics matching, over three seeds and with perceptual (LPIPS, DISTS) and texture (R_H) metrics (Fig. 1).

Our contributions are: (i) a controlled comparison of ten configurations of the perceptual loss, which finds no network best on every metric (Table 1); (ii) evidence that local statistics matching brings DINOv2-S grain closer to normal dose at no detectable PSNR cost; and (iii) a training-free screen of the networks, which does not predict the trained result (Fig. 2, Fig. 4).

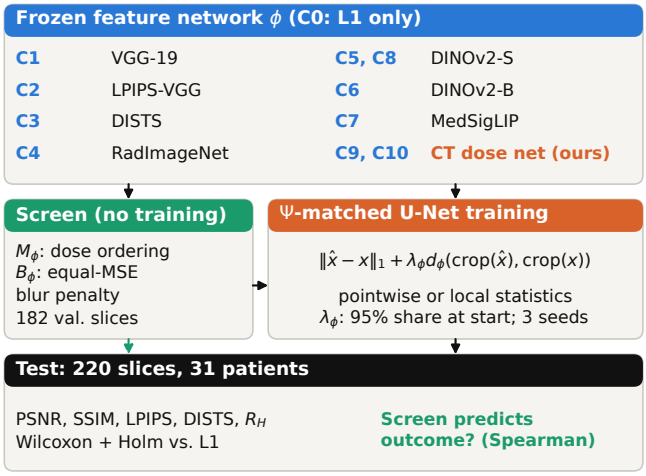


Fig. 1. Only the feature network in the loss changes; its weight is matched by influence, and a screen scores each network before training.

2. METHOD

Objective. A U-Net f_θ maps a low-dose slice to $\hat{x}$ and is trained against the normal-dose slice x with

$$\mathcal{L}_\phi = \|\hat{x} - x\|_1 + \lambda_\phi \, d_\phi\big(\mathrm{crop}(\hat{x}), \mathrm{crop}(x)\big),$$

where crop is one random 224×224 crop shared by prediction and target and d_ϕ a distance in the features of a frozen network ϕ. With $\bar{\phi}_l$ the channel-normalized features of block l (C_l channels, $H_l \times W_l$ positions), pointwise matching is

$$d^{\mathrm{pt}}_\phi(\hat{x}, x) = \sum_l \frac{1}{H_l W_l} \sum_{h,w} \big\| \bar{\phi}_l(\hat{x})_{hw} - \bar{\phi}_l(x)_{hw} \big\|_2^2. \qquad (1)$$

CT noise is random, so the output cannot copy the reference grain pixel by pixel. Local statistics matching asks only that each 4×4 window $\omega \in \Omega_l$ of each channel has the same feature mean μ and standard deviation σ:

$$d^{\mathrm{st}}_\phi(\hat{x}, x) = \sum_l \frac{1}{C_l |\Omega_l|} \sum_{c,\omega} \big[1 - s(\hat{\mu}, \mu, c_1)\, s(\hat{\sigma}, \sigma, c_2)\big], \qquad (2)$$

Table 1. Every perceptual loss costs PSNR against L1; the best LPIPS and DISTS values lie within one seed spread of each other, and statistics matching gives DINOv2-S its best texture. Test split (220 slices, 31 patients, 100k). Values: mean±std over seeds 42, 123 and 456. Bold: best trained configuration. † significantly better and ‡ significantly worse than C0 (L1) (paired Wilcoxon on the 31 patient means of seed-averaged slices, Holm over 30 tests, p < 0.05; PSNR, LPIPS and $|1 - R_H|$ only). * scored with its own training metric. point.: pointwise, stat.: local statistics matching.

ID	Network	Matching	PSNR (dB) ↑	SSIM ↑	LPIPS ↓	DISTS ↓	R_H (→1)	$\lvert 1 - R_H \rvert$ ↓
	LDCT input		23.04	0.7423	0.2843	0.1641	6.174	5.275
C0	–	–	**30.92±0.07**	**0.8691±0.0025**	0.1043±0.0010	0.1112±0.0006	0.657±0.026	0.352±0.026
C1	VGG-19	point.	30.32±0.22‡	0.8606±0.0022	0.0592±0.0010†	0.0668±0.0023	0.798±0.035	0.226±0.030†
C2	LPIPS-VGG	point.	30.39±0.31‡	0.8630±0.0021	0.0563±0.0045$^{\dagger *}$	**0.0645±0.0024**	0.777±0.012	0.242±0.009†
C3	DISTS	stat.	29.81±0.20‡	0.8326±0.0244	**0.0557±0.0066**†	0.0664±0.0014*	0.820±0.014	0.222±0.018†
C4	RadImageNet	point.	28.70±0.32‡	0.8259±0.0075	0.0868±0.0055†	0.0847±0.0009	0.798±0.040	0.286±0.017†
C5	DINOv2-S	point.	30.10±0.51‡	0.8625±0.0063	0.0629±0.0078†	0.0707±0.0068	0.682±0.046	0.333±0.044†
C6	DINOv2-B	point.	30.20±0.44‡	0.8618±0.0044	0.0614±0.0027†	0.0713±0.0028	0.731±0.027	0.291±0.029†
C7	MedSigLIP	point.	29.84±0.39‡	0.8596±0.0061	0.0750±0.0114†	0.0737±0.0070	0.531±0.063	0.477±0.060‡
C8	DINOv2-S	stat.	30.26±0.28‡	0.8614±0.0068	0.0579±0.0112†	0.0654±0.0047	0.857±0.017	0.195±0.013†
C9	CT dose net v2	point.	30.44±0.10‡	0.8514±0.0219	0.0673±0.0031†	0.0973±0.0002	0.850±0.018	0.203±0.010†
C10	CT dose net v2	stat.	29.80±0.19‡	0.8475±0.0039	0.0721±0.0063†	0.0982±0.0036	0.951±0.058	**0.167±0.026**†

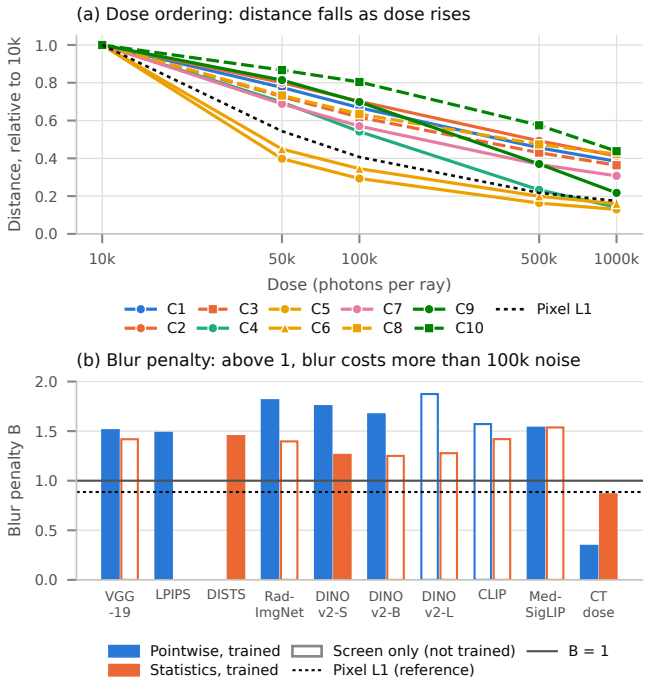


Fig. 2. Training-free screen (validation split). Every pretrained network orders doses almost perfectly (a); only the CT dose network penalizes blur less than noise of the same pixel error (b).

with $s(a, b, c) = (2ab + c)/(a^2 + b^2 + c)$ and $c_1 = c_2 = 10^{-6}$: the texture term of DISTS [11], applied to any network.

Feature networks (Table 2): VGG-19 at relu3_2 [2]; LPIPS-VGG [12] and DISTS with their own weights; RadImageNet ResNet-50, layers 1 to 3 [13]; DINOv2-S and DINOv2-B, blocks 3, 6, 9 [14]; MedSigLIP-448, layers 7, 14, 20 [15]. DINOv2-L and CLIP ViT-B/16 [16] enter the screen only. C8 and C10 use (2), C1, C4 to C7 and C9 use (1).

CT dose network. A loss network should change its features when the noise level changes. Three convolutional blocks (32, 64, 128 channels; 287,328 parameters) with a regression head predict the standardized $\log_{10}$ photon count of 128×128 training-split patches (normal dose fixed at 7), trained with smooth L1 for 10 epochs (lowest validation loss kept) and frozen; its block outputs are the features. This version replaced a six-class classifier after a validation screen; the adoption threshold on M_ϕ was lowered from 0.9 to 0.8 after seeing the screen, and the runs of the first version are an ablation in the repository. Unlike SACNN [3], its features are trained on dose alone.

Table 2. Feature networks. Every pretrained network orders doses (M ≥ 0.95); only the CT dose network has B < 1. † feature extractor of the CT dose network v2 (three conv blocks). λ: frozen loss weight. M: dose ordering, B: blur penalty of the training-free screen.

ID	Network	Matching	Params (M)	λ	M ↑	B
C1	VGG-19	point.	1.15	4.699	1.000	1.51
C2	LPIPS-VGG	point.	14.71	9.162	0.998	1.48
C3	DISTS	stat.	14.71	6.984	0.984	1.45
C4	RadImageNet	point.	25.56	2.211	0.954	1.81
C5	DINOv2-S	point.	22.06	1.634	0.998	1.75
C6	DINOv2-B	point.	86.58	2.497	0.997	1.67
C7	MedSigLIP	point.	428.57	3.755	0.999	1.53
C8	DINOv2-S	stat.	22.06	1.966	0.997	1.26
C9	CT dose net v2	point.	0.29†	1.549	0.971	0.34
C10	CT dose net v2	stat.	0.29†	2.802	0.852	0.86

Matched influence and screen. As in [10], we set $\lambda_\phi = 19\, S_1/S_\phi$, so the perceptual term is 95% of the loss at initialization (S_1, S_ϕ: summed L1 and feature terms of the untrained U-Net). Before training, each network is screened on the 182 validation slices: the dose ordering M_ϕ is the mean Spearman correlation between $d_\phi(x^{(d)}, x)$ and $\log(1/d)$ over five photon counts d from 10^4 to 10^6, and the blur penalty B_ϕ is the summed distance of a Gaussian blur divided by that of the 10^5 noise at equal MSE. $B_\phi < 1$ means a denoiser can lower the loss by smoothing. Every network is trained regardless of its score.

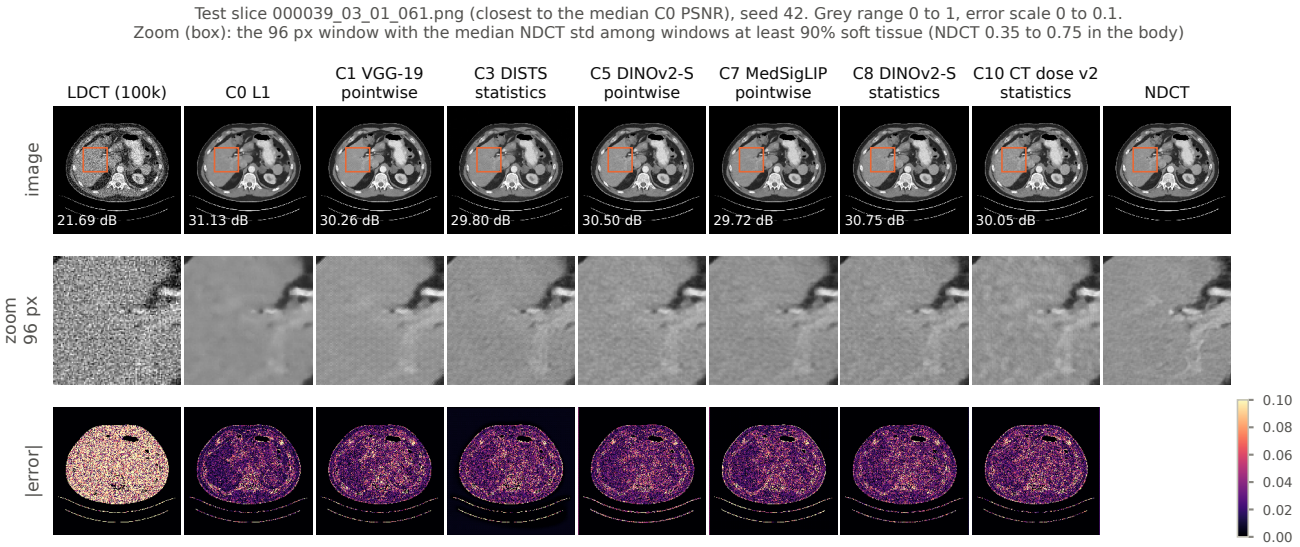


Fig. 3. Test slice closest to the median C0 PSNR, seed 42. Rows: image, zoom, $|\hat{x} - x|$; grey range $[0, 1]$, error scale 0 to 0.1. Zoom: the 96×96 window with the median NDCT standard deviation among windows that are at least 90% soft tissue (NDCT 0.35 to 0.75 inside the body mask). L1 smooths the grain away; the perceptual losses keep it.

3. EXPERIMENTS AND RESULTS

Setup. I2I 2025v1 slices from DeepLesion [17] (512×512, $[0, 1]$) with Poisson-simulated low-dose versions [18]; training input 10^5 photons; patient-level split of 899, 182 and 220 slices (115, 34, 31 patients). All eleven configurations (C0 is L1 only) train the same U-Net [19] with AdamW (10^{-4}), batch 8, 30 epochs and mixed precision, keeping the checkpoint with the lowest validation L1. Seeds 42, 123 and 456 vary the denoiser training only; λ_ϕ and the CT dose network are fixed. Code: `https://github.com/UnkindledTwo/perceptual_backbone_ldct`. Validation L1 in eval mode spikes in some runs (up to 8 of 30 epochs), so checkpoint selection on it is noisy; one run (C8, seed 42) has no recorded code revision. Metrics: PSNR, SSIM, LPIPS (AlexNet) [12], DISTS, and the high-band noise power ratio R_H at 0.25 to 0.5 cycles per pixel [20], tested as $|1 - R_H|$. Family A compares C1 to C10 with C0, Family B C8 with C5, C10 with C9 and C6 with C5, on PSNR, LPIPS and $|1 - R_H|$ (Holm over 30 and 9 tests), with paired Wilcoxon tests over the 31 test patients (seeds averaged per slice, slices per patient), effect size r and seed spread; slice-level tests are in the repository. Screen-versus-outcome Spearman correlations are exploratory: C5 and C8, C9 and C10, and C1 to C3 share feature networks.

Screen. Every pretrained network orders the doses almost perfectly (M_ϕ 0.93 to 1.00; Fig. 2), so ordering does not separate them. Among the feature networks, only the CT dose network has $B_\phi < 1$ (0.34 pointwise, 0.86 statistics), that is, it penalizes blur less than noise of equal MSE. Statistics matching lowers B_ϕ for every pretrained network except MedSigLIP (1.535 to 1.537) but raises it for the CT dose network.

Table 1. Against L1, every configuration loses PSNR (patient medians -0.50 to -2.26 dB, Holm $p < 10^{-4}$, $|r| \geq 0.93$, seed spread 0.1 to 0.5 dB) and lowers LPIPS (all ten significant, Holm $p \leq 0.021$). L1 keeps the best PSNR and SSIM. C1 to C3 (VGG-based) and C8 (DINOv2-S, statistics) form the best LPIPS and DISTS group, within one seed spread of each other (DISTS 0.065 to 0.067), and C10 (CT dose network, statistics) has the best $|1 - R_H|$ (0.168 against 0.352 for L1). Medical pretraining does not help: C4 (RadImageNet, pointwise) has the worst PSNR and LPIPS of the perceptual losses, and C7 (MedSigLIP, pointwise) is the only configuration with worse texture than L1 ($|1 - R_H|$ 0.477, R_H 0.53, smoother). The CT dose network matches grain but stays far behind on DISTS (0.097 to 0.098).

Fig. 3 and Fig. 4. L1 smooths the grain away (Fig. 3). Statistics matching brings DINOv2-S grain closer to normal dose: $|1 - R_H|$ falls from 0.333 for C5 (DINOv2-S, pointwise) to 0.195 for C8 (Holm $p = 4.9 \times 10^{-4}$, $r = -0.77$, seed spread 0.04), with no detectable PSNR or LPIPS change. The screen does not predict the outcome: B_ϕ correlates with $|1 - R_H|$ at $\rho = 0.84$ ($p = 0.004$; 0.71 without C2 and C3), the opposite sign to its premise, probably because the statistics configurations have both low B_ϕ and good texture. Size brings no reliable gain: DINOv2-B and DINOv2-S differ in neither PSNR nor LPIPS, and their texture difference, though significant when pooled, changes sign across seeds. Within pointwise configurations, more parameters even go with worse texture ($\rho = 0.89$, post hoc), consistent with [7].

4. CONCLUSION

Perceptual losses trade PSNR for perceptual quality, and no feature network is best on every metric. ImageNet VGG remains a sound default. Local statistics matching improves DINOv2-S grain but costs PSNR for the CT dose network, so it should be checked per network. Our training-free screen does not predict the trained result, so a loss network should not be chosen by such a screen alone. Limitations: simulated low-dose data from one dataset, one training dose and one denoiser. Next are real low-dose data and a CT feature network trained to detect structural distortion rather than dose.

5. COMPLIANCE WITH ETHICAL STANDARDS

This research study was conducted retrospectively using human subject data made available in open access by the NIH Clinical Center

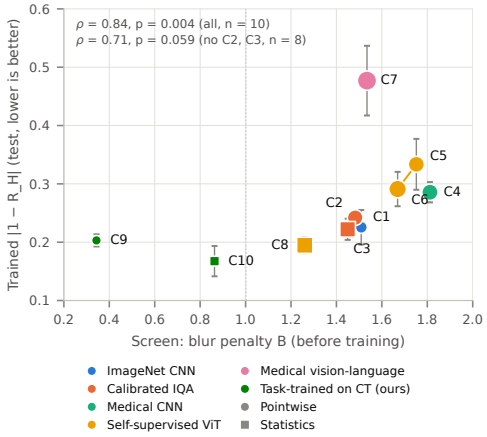


Fig. 4. Blur penalty against trained texture error. The screen does not predict the trained result ($\rho = 0.84$, exploratory).

(DeepLesion). Ethical approval was not required as confirmed by the license attached with the open access data.

6. ACKNOWLEDGMENTS

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